pH and Texture Composite (30 cm) Sampling
Supporting Document `IndigoCarbon_US-1_2023_0003`

June 27, 2024

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1 Context

When soil sampling for Indigo Carbon, one fifth (samples collected October 1, 2019 – March 15, 2020) or one third (samples collected March 16, 2020 – present) of carbon sampling locations are designated as a Bulk Density sampling locations (see `IndigoCarbon_US-1_2023_0004` for an explanation of the sample point prescriptions).

At each Bulk Density site, three different sample types are collected (Figure 1) and the sample type bolded below must be collected according to the instructions in this document.

- An individual 30 cm carbon sample according to the instructions in `IndigoCarbon_US-1_2023_0001`
- An individual 30.48 cm bulk density sample according to `IndigoCarbon_US-1_2023_0002`
- **A core that will contribute to a composite sample for pH and texture testing according to the work instructions below (applicable only to sites sampled March 16, 2020 – present)**

2 Inventory checklist

The inventory check list for this protocol is the same as for the Soil Carbon (30 cm) Sampling protocol (`IndigoCarbon_US-1_2023_0001`), with the addition of an extra bucket to hold the composite sample.

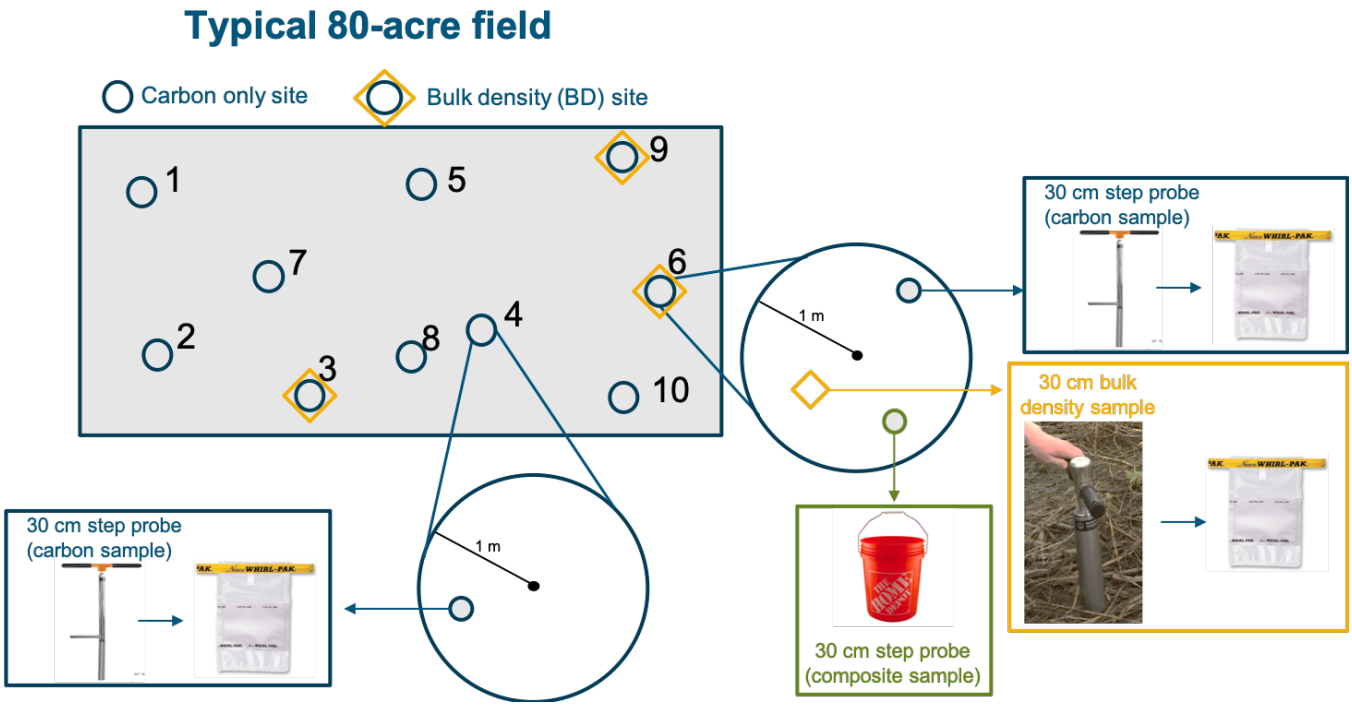


Figure 1: Indigo Carbon soil sample collection

3 Preparation

- Prepare sampling equipment before visiting the field. Review inventory checklist and make sure you have all necessary materials.
  - Take note of supplies in short inventory and request additional consumables from your Equipment Manager.
- All equipment can be stored in the trailer.
- Make sure you are familiar with all material in Indigo's field safety protocol. For this procedure you will need:
  - Safety Glasses
  - Steel Toe Shoes
  - Cut-Resistant Gloves
- Navigate to the appropriate sampling location by following the instructions outlined in `IndigoCarbon_US-1_2023_0004`.
  - There are specific instructions in `IndigoCarbon_US-1_2023_0004` that will guide you when faced with situations when you are unable to navigate to the prescribed point due to any reason (flood, fence, private property) or find that the point is in a compromised sampling location (field edge, access road, rock outcropping).
- Clear residue or vegetation from the soil surface where you intend to take a soil core. Only sample bare mineral soil and avoid residue in the sample (Figure 2).

4 Sampling procedure at each bulk density sampling location

- When you arrive at a sampling location that has been designated as a bulk density sampling site:
  - Collect an individual 30 cm carbon sample according to `IndigoCarbon_US-1_2023_0001`.
  - Collect an individual 30.48 cm bulk density sample according to `IndigoCarbon_US-1_2023_0002`.



Figure 2: Residue clearing to prepare sample core extraction

- After the individual samples have been bagged and logged into Indigo Fields v2.0 (IF 2.0), collect an additional 30 cm core using either the step probe or drill auger according to `IndigoCarbon_US-1_2023_0001`.
  - **Instead of bagging this 30 cm sample and logging it into IF, simply add it to the bucket designated to hold the composite sample**.
  - **NOTE: Whenever possible, the core contribution to the texture/pH composite sample should be the same depth as the carbon sample collected at the same point. If faced with a scenario where this is not achievable, the deeper of the two cores should be designated as the carbon sample.**
- Proceed with sampling. When you arrive at the next bulk density sampling location, repeat the steps above, **placing the new composite core into the same bucket with the core from the first bulk density sampling location**.
- This process should be repeated until all samples within a field have been collected.
  - When the field has been fully sampled, the total number of cores in the composite bucket should equal the number of bulk density sites on the field.

5 Subsampling of the composite

5.1 Subsampling procedure

- When you have finished sampling all locations within a field, fully mix all cores within the composite bucket with your hand or a trowel so that the entire sample is homogeneous.
- Transfer ~2 cups of the homogenized sample to a 27 oz Whirl-Pak and attach a QR code to the front of the bag.
- Close Whirl-Pak by flipping the bag twice and closing the tabs to ensure the bag is completely sealed. Label the outside of the bag with the grower, farm, and field information as well as the label 'texture + pH'.

5.2 Scan composite sample into IF

- In the mobile app Indigo Fields (Figure 3), you should record information about each texture core at the respective *points*, and then scan the full composited sample at the field level after you are done sampling the field and have made the composite sample (Figure 3).
  - At each point that requires a texture core to be pulled, there will be questions you have to fill out, including:
    - "Are you able to take this sample".
    - A question about sample depth.

    - A question about sampling method.
  - After you have completed sampling for the field, and you have composited the sample in one bag with a QR code, you should scan the Texture/pH soil sample QR code under the Survey Completion section of the task.
    - This is required before you can submit the survey.

6 Fields with Less Than 3 Bulk Density Sampling Locations May 2, 2020 – February 28, 2021

- Texture and pH require a minimum sample volume. In cases where there are only one or two bulk density sampling sites, the 'texture/pH composite' would consist of only one or two single step probe/augered samples. This will not be enough volume.
- To account for this, in small fields with one or two bulk density sites please use the following guidelines:
  - Fields with 2 bulk density sites = pull two 30 cm cores at each bulk density site for composite.
  - Fields with 1 bulk density site = pull three 30 cm cores around that single bulk density site for composite.
  - You will likely need to bag the entire composite sample to have the requisite 2 cups of material.

7 Fields with Less Than 3 Bulk Density Sampling Locations or Short Cores (March 1, 2021 - Present)

- Texture and pH analyses require a minimum sample volume. In cases where only one or two bulk density sampling sites are able to be sampled or full depth samples cannot be collected, the standard 'texture/pH composite' protocol would likely not produce enough dry soil mass.
- To account for this, in fields with 1-2 bulk density sites or short sample cores, please use the following guidelines. **Carbon samples referred to below represent the companion carbon samples collected at each bulk density site along with bulk density and texture/pH composite sub-samples.**
  - Fields with > 4 bulk density sites:
    - No adjustment to protocol required
  - Fields with 3-4 bulk density sites:
    - If all carbon samples are collected to 30 cm: No adjustment to protocol required.
    - If > 1 carbon sample(s) are collected < 30 cm: Pull two cores at each bulk density site equal to the sample depth achieved for the carbon sample at the site for composite.
  - Fields with 2 bulk density sites:
    - If all carbon samples are collected to 30 cm: Pull two 30 cm cores at each bulk density site for composite.
    - If ≥ 1 carbon sample(s) are collected < 30 cm: Pull three cores at each bulk density site equal to the sample depth achieved for the carbon sample at the site for composite.
  - Fields with 1 bulk density site:
    - If the carbon sample is collected to 30 cm: Pull three 30 cm cores around that single bulk density site for composite.
    - Sample(s) collected < 30 cm: Pull four cores around that single bulk density site equal to the sample depth achieved for the carbon sample for composite.
    - **NOTE: Sampler will likely need to bag the entire composite sample to have the requisite 2 cups of material.**

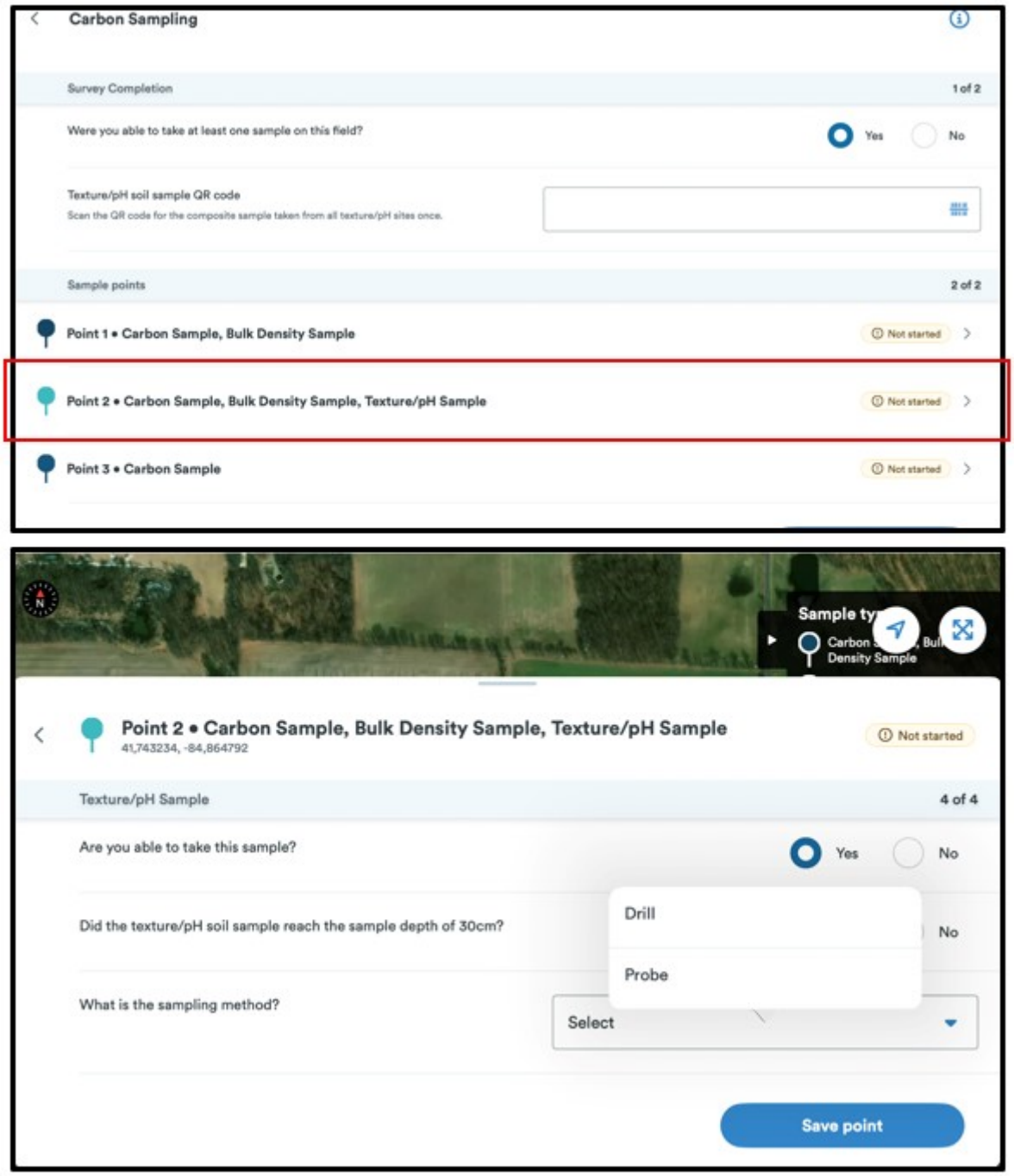


Figure 3: Selecting and recording a ph/texture sample in IF 2.0

8 Summary of Changes

Section	Date	Summary of Changes
Throughout	May 12, 2024	Changes made to refrences made to legacy version if Indigo Fields Mobile